

سلول و ملکول عملی

1- Flame Photometry is for determining :

- ☐ Concentration of colored solutions
- ☒ Concentration of metal ions
- ☐ Concentration of liquids
- ☐ Concentration of organic solutions

2- In a titration experiment how do we know the end point has reached?

- ☐ With an acid
- ☐ Measuring pH
- ☐ With a base
- ☒ Using indicators

3- Which of the following is insoluble in water?

- ☐ Fructose
- ☒ Starch
- ☐ Glucose
- ☐ Sucrose

4- Which of the following tests is specific for amines, phenols and imidazole?

- ☐ Xanthoproteic test
- ☐ Biuret test
- ☐ Molisch test
- ☒ Paulys test

5- What is the color of Phenolphthalein in acidic solutions?

- ☒ Colorless
- ☐ Blue
- ☐ Pink
- ☐ Yellow

6- Which of the following is used in inorganic chemical analysis ?

- Colorimetry
- Spetrophotometry
- Chromatography
- ✓ Flame Photometry

7- Which of the following tests indicates the presence of a reducing sugar?

- Millons test
- Ninhydrin test
- Molisch test
- ✓ Fehling's test

8- Amax is defined as:

- ✓ coloured compound absorbs maximum light at a particular wavelength
- explains the transmittance of light
- it is useful for colorless compounds
- relationship between absorbance and concentration

9- Ninhydrin with evolution of CO₂ forms a

coloured complex with

- Histamine
- Serotonin
- Peptide bond
- ✓ α-Amino acids

10- In gel permeation chromatography separation is based on....

- ✓ Size of molecule
- Solubility
- Ion exchanging
- Affinity

11- Guanidine group of arginine gives

positive test with

- ✓ Sakaguchi reagent
- Molisch's reagent
- Trichloroacetic acid
- Lead acetate

12- Presence of carbohydrates is confirmed by which test?

- Biuret test
- Millons test
- ✓ Molisch test
- Ninhydrin test

13- Millon's test is for identification of

- Tryptophan
- Proline
- Arginine
- ✓ Tyrosine

14- The mobile phase in chromatography is :

- Solid-solid
- ✓ Liquid-gas
- Solid-gas
- Liquid-solid

15- In Mohr pipettes :

- Has a single graduation to deliver specific volume
- ✓ The graduation on these end before the tip
- The graduation marks continue to the tip
- Are disposable

16- Which of the following tests is specific for albumin?

- ☐ Xanthoproteic test
- ☒ Millon's test
- ☐ Molisch test
- ☐ Paulys test

17- Watch Glass is used to :

- ☐ heat and evaporate liquids
- ☒ dry and weigh solid compounds
- ☐ dissolve the solids
- ☐ crush solids into powders

18- Mortar and Pestle are used to :

- ☐ heat and evaporate liquids
- ☐ dissolve the solids
- ☐ dry and weigh solid compounds
- ☒ crush solids into powders

19- Which of the following is used to dispensing solid chemicals?

- ☒ Spatulas
- ☐ Forceps
- ☐ Glass Rods
- ☐ Mortar

20- Aromatic amino acids can be detected by

- ☐ Sakaguchi reaction
- ☒ Xanthoproteic reaction
- ☐ Millon-Nasse reaction
- ☐ Hopkins-Cole reaction

21- Hopkins-Cole test is for identification of

- Cysteine
- Arginine
- Tyrosine
- ✓ Tryptophan

22- Iodine solution produces no color with

- ✓ Cellulose
- Dextrin
- Starch
- Glycogen

23- Which of the following indicates the presence of amino acids?

- Molisch test
- ✓ Ninhydrin test
- Fehling's test
- Millons test

24- The purple ring of Molisch reaction is due

to

- ✓ Furfural + α Naphthol
- Furfural + H_2SO_4 + α -Naphthol
- Furfural
- $^{\circ}C$ Naphthol

25- A positive Benedict's test is not given by

- Maltose
- Lactose
- Glucose
- ✓ Sucrose

26- What is the color of Phenolphthalein in basic solutions :

Yellow

Colorless



Pink

Blue

27- All the following is true regarding the basic principle of spectrophotometer except ?

It depends on how much a chemical substance absorbs light

Its done over a certain range of wavelength



is one of the most useful qualitative analysis

It depends on how much a substance transmits light